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3D geometric features based real-time American sign language recognition using PointNet and MLP with MediaPipe hand skeleton detection

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ABSTRACT

This paper presents a robust and fast hand gesture recognition method. Utilizing the 21 hand key points predicted by MediaPipe developed by Google, we calculate additional 3D geometric features, including the angles of each finger joint. These geometric features are then subjected to deeper feature extraction using a Multilayer Perceptron (MLP). Simultaneously, these hand key points are normalized in 3D spatial space, and the finger and palm regions are interpolated to generate a point cloud representing the hand skeleton. Subsequently, PointNet is employed for feature extraction from this point cloud. The features extracted by MLP and PointNet are fused, and a classifier is employed to output the probability distribution of hand gestures for each frame of a 3D video stream. Furthermore, a probability machine is designed to analyze the probability sequence, providing the overall gesture prediction. In the experimental phase, we trained a model to classify American Sign Language (English letters and numbers 0 to 9). The results demonstrate an overall accuracy of 99.81 %. The entire method operates in less than 10 milliseconds (using an NVIDIA 2080Ti graphics card), showcasing its real-time detection capabilities.

1. Introduction

American Sign Language (ASL) is a language that communicates through signs, facial expressions, and body language, as shown in Fig. 1. It is one of the most commonly used forms of communication in the North American Deaf community. It can also be used in the field of human-robot interaction, where even people who don't understand ASL syntax can communicate commands to a robot by simply learning ASL's representation of English letters and numbers. This type of interaction is suitable for manipulating mobile robots in scenarios where it is not convenient to carry input devices (mouse, keyboard, or teach pendant).

In recent years, many methods for hand skeleton or landmarks detection have emerged [1–6]. Among them, it is worth mentioning that Mediapipe developed by Google [1], using the Mobilenet [7] as the backbone, realizes real-time 2D hand landmarks detection on mobile devices. We consider that using its hand skeleton prediction to further design a lightweight classifier to predict ASL gestures is a fast and simple approach. In contrast to our proposed approach, in Refs. [8–12] end-to-end methods for recognizing ASL gestures directly from images are presented. Obviously, these approaches require more extensive training periods. Because the convergence of the image feature extractor is much more time-consuming relative to a classifier. This shortcoming is revealed more clearly when the network is trained to classify more different sign languages.

In addition, gesture recognition by analysing 2D landmarks alone is not robust enough. Because these landmarks are merely projections of 3D points on a 2D plane, the analysis is underdetermined. In other words, only the rays about these 3D points are known. Junger et al. [14] discussed the problems caused by this underdetermination for subsequent image processing. Therefore, the introduction of a 3D points analysis is necessary. For this, besides 2D landmarks, Mediapipe also predicts 3D landmarks, but because of the above limitation, these 3D predictions are sometimes incorrect. With respect to this, in the references [15,16], the constraints on the hand joint angles have been studied in detail. With these constraints the 3D landmarks by Mediepipe could be corrected. In many advanced studies on human-robot interaction nowadays, there is often an emphasis on algorithmic performance while overlooking the human biology and physiological parameters. In Refs. [17,18], vision-based methods for measuring vital signs such as heart rate and oxygen saturation were introduced. Otherwise [19,20], showed the applications of human body temperature analysis in human-computer interaction. These studies demonstrated that combining algorithm with biomedical principles is effective.

To overcome these limitations in our method, 3D hand landmarks are first predicted with the help of Mediapipe. Then these landmarks are corrected according to the constraints of the hand joints. Subsequently, their spatial distribution is analyzed using PointNet [21], a fast global point cloud feature extractor. Meanwhile the joint angles that are computed in the process of correcting the landmarks are fed into a series of MLP layers as explicit intermediate features, which are identified as the local feature extraction of hand gestures. This setup ameliorates the error due to underdetermination as well as ensures the compactness and efficiency of the network. Certainly, no network can guarantee 100 % accuracy. Therefore, we designed a probabilistic machine that works like a Kalman filter for filtering noise.

2. Method

As mentioned above, Mediapipe is a fast hand landmarks detection method based on color image processing. It predicts 21 2D-landmarks

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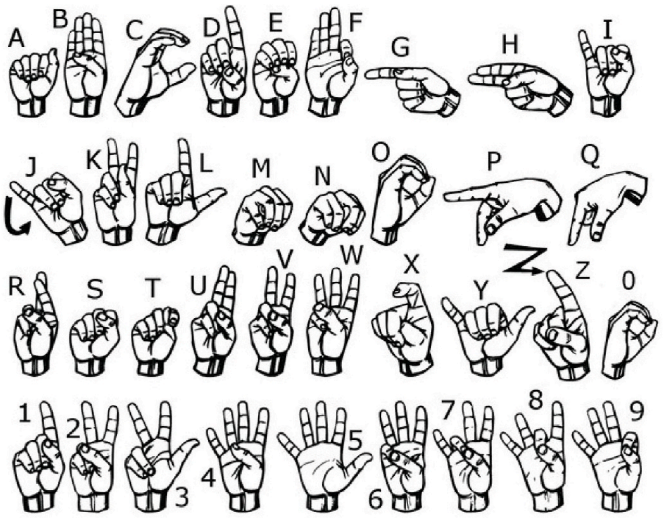


Fig. 1. American Sign Language (the gestures of English letters and numbers) [13].

while simultaneously providing the 3D coordinates of them. The x and y coordinates are normalized to range from 0 to 1, describing the relative positions of these points in the 2D image. The z coordinate is dependent on the values of x and y and is used to describe the orientation of the palm and fingers.

These 3D relative landmarks predicted exhibit errors occasionally. For example, the Fig. 2 (a) shows the prediction by Mediapipe for a hand gesture (fist). It is clear that the predicted bending of the index finger and pinky is not achievable by human because the rotation of finger joints is constrained (not $(-\pi, +\pi)$). Moreover, the joints should not be analogized to a 3-Degree-of-Freedom (DoF) ball-and-socket joint. Therefore, it is not rigorous to only rely on 21 points to describe hand gestures. Instead, the constraints should be utilized to mathematically model the kinematic of the hand, in order to correct the prediction by Mediapipe. For the joint angles, based on summarizing the results of [15, 16], we decided to use the constraints shown in Table 1 in our hand recognition approach.

2.1. Hand descriptor

In this section we present a hand descriptor of the hand, which is used to correct the false 3D landmarks. An intermediate local feature vector is also generated for each gesture.

We consider that each person has individual differences in the length

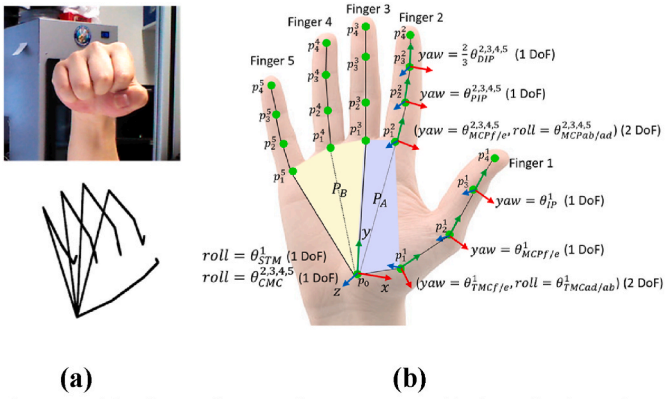


Fig. 2. (a) Above shows a fist gesture and below displays the 3D landmarks predicted by Mediapipe for this gesture; (b) Hand kinematics i.e., finger joints and their Degrees of Freedom (DoF).

Table 1
Description of constraints for each degree of freedom of each finger joint.

Fingers	Joints	yaw	roll
Thumb (Finger 1)	Scapahoid-Trapezium (STM)	–	−90°
	Trapeziometacarpal (TMC)	0°–60°	−45°-0°
	Metacarpophalangeal (MCP)	0°–45°	–
	Interphalangeal (IP)	0°–90°	–
Index (Finger 2)	Carpometacarpal (CMC)	–	−5°
	Metacarpophalangeal (MCP)	−30°-90°	−30°-30°
	Proximal interphalangeal (PIP)	0°–110°	–
	Distal interphalangeal (DIP)	0°–80°	–
Middle (Finger 3)	Carpometacarpal (CMC)	–	5°
	Metacarpophalangeal (MCP)	−30°-90°	−30°-30°
	Proximal interphalangeal (PIP)	0°–110°	–
	Distal interphalangeal (DIP)	0°–80°	–
Ring (Finger 4)	Carpometacarpal (CMC)	–	10°
	Metacarpophalangeal (MCP)	−30°-90°	−30°-30°
	Proximal interphalangeal (PIP)	0°–120°	–
	Distal interphalangeal (DIP)	0°–80°	–
Little (Finger 5)	Carpometacarpal (CMC)	–	15°
	Metacarpophalangeal (MCP)	−30°-90°	−30°-30°
	Proximal interphalangeal (PIP)	0°–135°	–
	Distal interphalangeal (DIP)	0°–90°	–

of finger bones (phalanges and metacarpal bones), and these differences should not be regarded for gesture classification. Therefore, we first normalized the 3D hand skeleton predicted by Mediapipe, keeping the hand joint angle constant, and convert the hand bone lengths to the average lengths introduced in Ref. [22].

Same as the finger bone length, the recognition of gestures does not depend on the position of the palm in the camera coordinate system. However, because some differences between gestures in ASL are solely due to variations in palm orientation (as shown in Fig. 1: D and 1, V and 2), the orientation should be calculated by

$$\begin{aligned} \vec{z} &= \widehat{p_1^1 p_0} \times \widehat{p_1^2 p_0} \\ \vec{x} &= \widehat{p_1^1 p_0} \times \vec{z} \\ \vec{y} &= \vec{z} \times \vec{x} \\ r &= [\vec{x}^T, \vec{y}^T, \vec{z}^T], \end{aligned} \tag{1}$$

where $\widehat{p_1^1 p_0}$ and $\widehat{p_1^2 p_0}$ are two unit vectors, and they denote the directions from the wrist point p_0 to the middle and index fingers respectively, as shown in Fig. 2 (b). Here we determine the basis vector of the Z-axis $\vec{z}$ using the metacarpal bones of these two fingers (perpendicular to both of them). Then a vector perpendicular to both the middle finger and $\vec{z}$ is defined as the basis vector of X-axis $\vec{x}$. The basis vector of Y-axis $\vec{y}$ can be calculated using $\vec{z} \times \vec{x}$ simply. Finally, the rotation matrix is formed by this set of orthogonal basis.

Because in our definition, the PIP joint angles are in the interval [0, π] (without considering constraints), we use the dot product and arccos () to solve for them. The other joint angles (interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$) with one DoF is solved using the cross product and arcsin(). They can be calculated by the following equation simply.

$$\theta_j^i = \begin{cases} arccos\left(\vec{y} \cdot \widehat{p_{j+1}^i p_j^i}\right) & i = 2 \\ arc\,sin\left(\left\|\vec{y} \times \widehat{p_{j+1}^i p_j^i}\right\|\right) & i \neq 2 \end{cases} \tag{2}$$

In equation (2), $\vec{y}$ denotes the basic vector of y-axis in the local coordinate system of the joint θ_j^i. For joints with two DoF (the Fig. 4 shows an example about joint MCP of the index finger), we introduce an intermediate point $p_{j+1}^{i\,\prime}$, which is the projection of the point p_{j+1}^i onto the YZ-plane in the local coordinate system of the joint θ_j^i. We can calculate θ_{j-yaw}^i and θ_{j-roll}^i by using the equation (3). Finally, all the joint angles will be subjected to the constraints in Table 1.

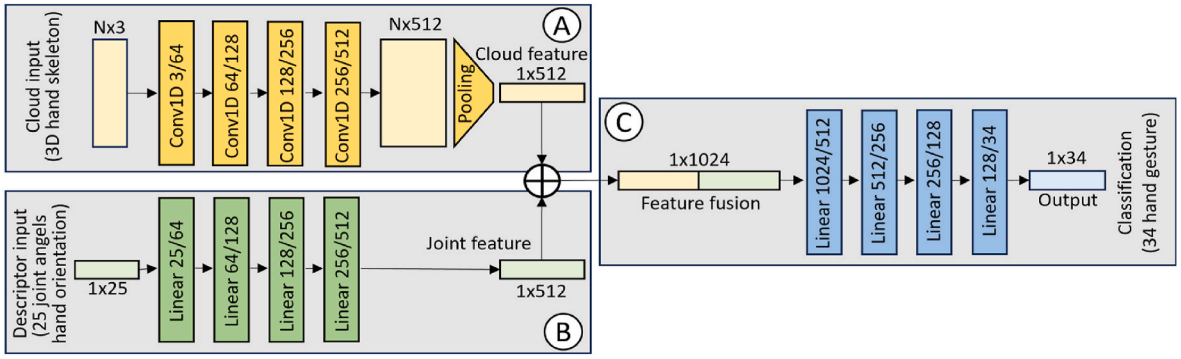


Fig. 3. The neural network for classifying ASL-gestures. It consists of three components, A: a point cloud feature extractor for encoding the hand skeleton point cloud; B: a set of MLP for encoding the hand descriptor; C: feature fusion and classifier.

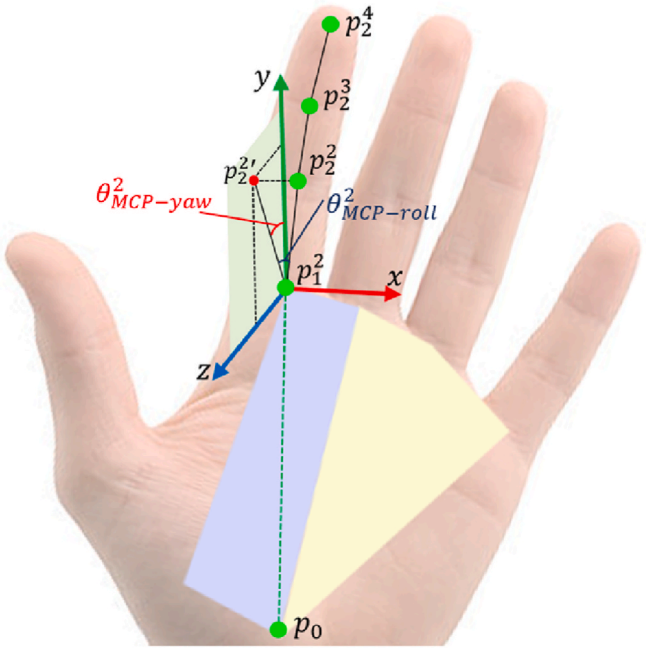


Fig. 4. The description of a joint with two DoF (MCP-joint of index finger) and the intermediate point for calculation joint angels.

$$\theta^i_{j-yaw} = arc\ sin\left(\left\|\overrightarrow{y} \times \widehat{p^i_{j+1}{}' p^i_j}\right\|\right)$$
$$\theta^i_{j-roll} = arc\ sin\left(\left\|\widehat{p^i_{j+1}{}' p^i_j} \times \widehat{p^i_{j+1} p^i_j}\right\|\right) \quad (3)$$

2.2. Network

The Fig. 3 demonstrates a network designed for static hand gesture classification, composed of three modules:

2.2.1. Point cloud feature extractor

This module is utilized for feature extraction of the hand skeleton point cloud computed from the hand descriptor (constrained joint angles and average lengths of phalanges and metacarpal bones in Ref. [22]). PointNet is a lightweight network, capable of rapid training and inference. However, its limitation lies in its ability to only extract global features from point clouds. Nonetheless, in our application, the extraction of local features from the point cloud is unnecessary. Therefore, an architecture similar to PointNet is employed in this module. As mentioned above, palm orientation is an important characteristic in distinguishing ASL gestures. Thus, this point cloud feature extractor should possess scale invariance (constant bones length) and translation invariance (constant hand position), but not rotational invariance. This differs from most applications of point cloud processing. Therefore, the T-Net module of the standard PointNet architecture, which ensures rotational invariance, should be discarded.

2.2.2. Joint feature extractor

This module is utilized for explicit feature extraction of the joints (25 joint angles and hand orientation i.e., 3 Euler angles) and consists of several linear layers (MLP). We consider these joint states to be the crucial local features distinguishing gestures.

2.2.3. Feature fusion and classifier

Module A and B respectively extract a 512-dimensional feature vector. These vectors are simply concatenated and fed into a classifier consisting of several linear layers (MLP) to predict the probabilities of 34 gesture classes.

2.3. Probability machine

The utilization of the above-mentioned network for gesture

prediction inevitably leads to some misclassifications. Furthermore, even if the network's predictions are accurate, it does not guarantee that the gesture necessarily corresponds to the command the user intends to convey to the robot. For example, during the transition from gesture E to gesture B, the network might predict gesture C (as shown in Fig. 1). Although this prediction may be accurate at this brief moment, it lacks meaningful significance. Hence, we consider, the prediction of the network should only be defined as observe event. A probabilistic machine is developed to serve as the final judgment for gesture recognition.

Firstly, we establish that one hand can only express one gesture at any given moment. For each gesture of ASL at time t_i, the probability p_{i-1} at time t_{i-1} is defined as the prior probability. The probability o_i of this gesture being predicted by the network at time t_i is defined as an observed probability. We define that the probability p_i can be calculated using the equation

$$p_i = p_{i-1} + o_i I(p_{i-1}, \Delta t) + (1 - o_i) D(p_{i-1}, \Delta t), \tag{4}$$

where $I()$ and $D()$ are probability update functions representing the growth and decay terms, respectively. They depend on p_{i-1} and the time difference $\Delta t = t_i - t_{i-1}$. We aim for an accelerated increase in the probability of a gesture if it is continuously observed to occur. Conversely, the probability should decay at an accelerated rate. Obviously, power functions and exponential functions both satisfy this characteristic, whereas linear functions do not.

The top part of Fig. 5 shows the ideal growth by the power and exponential functions where the probability sequence starts at time s_0 with $p_0 = 0$ and continuously observed probability $o_i = 1$. The bottom part depicts the ideal decay where the probabilities start with $p_0 = 1$ and continuously observed probability $o_i = 0$. However, in practice, it is not possible to have such an ideal observed probability. Usually, it contains Gaussian noise ($o_i < 1$) as well as the possibility of misjudgment. It is proved through our experiments that the optimal settings are when using an exponential growth function and a power decay function (E + P):

$$\begin{aligned} I_{exp}(p_{i-1}, \Delta t) &= \left(2^{\frac{\Delta t}{a}} - 1\right)(p_{i-1} + 1) \\ D_{pow}(p_{i-1}, \Delta t) &= -2\left(\left(\frac{\Delta t}{a}\right)^2 + \frac{\Delta t}{a}\sqrt{1 - p_{i-1}}\right), \end{aligned} \tag{5}$$

where a denotes a sensitivity factor of this probability machine. Its physical interpretation is the duration when the probability grows from 0 to 1 (or decays from 1 to 0) in the ideal case in Fig. 5. The power update function has a slope of 0 at extremities ($p_{i-1} = 0$ *or* $p_{i-1} = 1$), with increments and decrements containing only second-order terms, which are very low values. This endows it with excellent misjudgment resilience. However, this also results in the more delay in observation sequences under the same sensitivity relative to exponential update function. In the presence of severe Gaussian noise, the delay is amplified. Therefore, the choice of an exponential growth function is to allow the system to respond more quickly to newly emerging gestures. The selection of a power decay function is to endow the system with resistance to misjudgments during the continuous process of gestures.

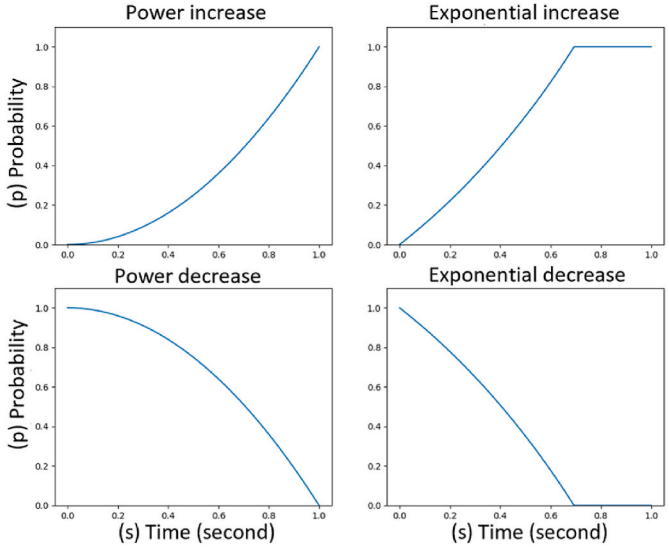


Fig. 5. The ideal probability growing and decaying with the observed probability $o_i = 1$ and $o_i = 0$.

3. Experiments

3.1. Results of classification network

To validate the robustness of this network, we enlisted the help of four volunteers for collecting the dataset. None of them had prior experience with ASL. Volunteers V1 and V2 diligently studied ASL with the help of an instructional video, while volunteers V3 and V4 learned ASL only through visual demonstration in Fig. 1 and verbal guidance from V1 and V2. Approximately 200 samples were collected for each gesture by each volunteer. Thus, we obtained four sub-datasets from these four testers.

We employed the k-fold cross-validation method. First, we independently selected each subset as a training set to train the network and validate the other three subsets. Then, three subsets were selected as the training set to validate the remaining dataset. For each training, we used a learning rate of $lr = 0.001$ for 10 epochs, followed by training with a learning rate of $lr = 0.0001$ for 30 epochs. The experimental results are shown in the Table 2.

In this table, the columns represent different training sets, while the rows represent different validation sets. Each cell indicates the results achieved by a specific training set when evaluated on a particular validation set. Those grids with 100 % accuracy can be ignored as they represent identical results between the training set and the validation set, and they only serve to demonstrate that the network can be converged. Moreover, it can be observed that using data from only one tester as the training set is not sufficient. The results demonstrate a significant improvement of robustness when training on three subsets as the training set. But the results of the configuration (train-V1V2V3/valid-V4) remain low. Therefore, we attempted data augmentation by introducing random noise during training. We added Gaussian noise with standard deviations of 10° for hand orientation and 5° for joint angles. For this training batch, we used $lr = 0.001$ for 30 epochs, followed by training with $lr = 0.0001$ for 50 epochs. Table 3 presents the results.

The results indicate that data augmentation leads to approximately a 5 %–10 % improvement for models trained on a single subset. In the cases of training with three subsets, there is an approximately 3 % improvement with the configuration (train-V1V2V3/valid-V4), while there is almost no impact on the other three configurations. Finally, we merged four datasets and divided them into training and validation sets

Table 2
The ASL classification results using k-fold cross validation without data augmentation.

train\valid	V1	V2	V2	V3
V1	100 %	**84.12 %**	**84.09 %**	**65.88 %**
V2	**73.67 %**	100 %	**70.36 %**	**61.32 %**
V3	**60.92 %**	**67.81 %**	100 %	**54.78 %**
V4	**80.16 %**	**83.82 %**	**85.80 %**	100 %
V2,V3,V4	**92.56 %**	100 %	100 %	100 %
V1,V3,V4	100 %	**94.57 %**	100 %	100 %
V1,V2,V4	100 %	100 %	**96.33 %**	100 %
V1,V2,V3	100 %	100 %	100 %	**78.13 %**

Table 3
The ASL classification results using k-fold cross validation with data augmentation.

train\valid	V1	V2	V3	V4
V1	100 %	**88.25 %**	**86.72 %**	**69.51 %**
V2	**79.16 %**	100 %	**73.22 %**	**72.85 %**
V3	**68.06 %**	**69.31 %**	100 %	**64.58 %**
V4	**79.70 %**	**91.71 %**	**94.70 %**	100 %
V2,V3,V4	**92.50 %**	100 %	100 %	**99.95 %**
V1,V3,V4	100 %	**94.81 %**	100 %	**99.98 %**
V1,V2,V4	100 %	100 %	**96.10 %**	**99.95 %**
V1,V2,V3	100 %	100 %	100 %	**81.02 %**

Table 4
The ASL classification results of dividing the entire dataset into training and validation sets in different proportions.

Train 50 % Validation 50 %	Train 25 % Validation 75 %	Train 10 % Validation 90 %
99.81 %	99.20 %	95.15 %

according to different proportions. Ensuring both datasets include data for all gestures by all testers. The train setup remains as $lr = 0.001$ for 10 epochs and $lr = 0.0001$ for followed 30 epochs. Table 4 presents these experimental results.

3.2. Probability machine results

In order to verify the performance of our probability machine, we first generated a set of simulated data that simulates a sequence of observed probabilities of a gesture over a period of time (0–15 seconds). As shown the blue curve in Fig. 6 (a), we set this gesture to occur within the intervals of t∈(1,3), t∈(5,8), t∈(10.75,11.25) and t∈(12,14) seconds. At the time t = 2 and t = 7, we added errors where the gesture was happened but not observed (misjudgment A). At the time t = 10, an error was added, where the observations falsely predicted the occurrence of the gesture (misjudgment B). Moreover, we added global Gaussian noise with std = 0.2. As shown in the top part of Fig. 6 (a), the power grow/decay functions barely react to the misjudgments at time t = 2, t = 7 and t = 10. However, its delay is large conversely, resulting in a brief gesture at time t = 11 is not detected. This is due to the Gaussian noise of 0.2 which causes power function to have a very low rate of change at the extremes. Therefore, we recommend using the E + P setting, as shown in the bottom part of Fig. 6 (a), which is most resistant to both misjudgments and Gaussian noise.

Furthermore, we also evaluated the probability machine using real data in practical. Fig. 6 (b) displays the ground truth of the hand gesture set of UVW (top), the observed probability sequences by network prediction (middle) as well as the response of the probability machine with the recommended update functions E + P (sensitivity coefficient a = 1) (bottom). In practical applications, the gestures of U and V are prone to confusion by the network. The results are in line with our expectations, with the final prediction of this hand gesture recognition system exhibiting excellent noise resistance.

4. Conclusions

The two main contributions of this work are: First, an easy-to-train and fast neural network is designed for gesture prediction. With the help of prediction of Mediapipe, by using the results of ergonomic

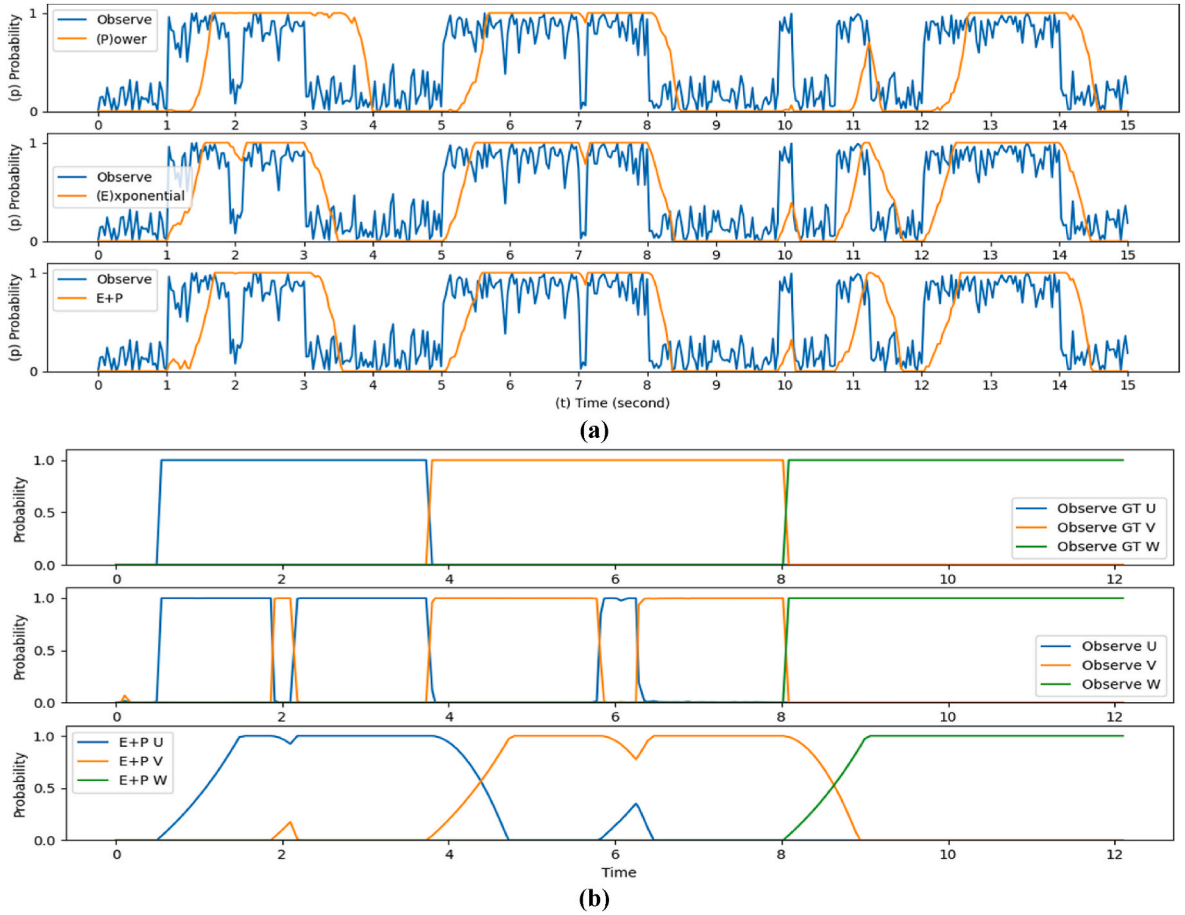


Fig. 6. Performance evaluation results for probability machine with simulated test data (a) and real test data (b).

studies as finger joint constraints to realize a more reasonable prediction of 3D landmarks of human hands. Moreover, in the classifier, global features based on point cloud analysis and local features according to finger joints are considered to achieve robust ASL classification. The robustness of our method to different individual samples is confirmed in the experiments by k-fold cross-validating the data from four testers. Second, we designed a novel probability machine to counteract the errors of neural network prediction. Experiments demonstrate that this probability machine has well resistance to misjudgments of less than 0.5 seconds (errors of gestures classification) and to Gaussian noise with std<0.2 (errors of observed probability). It is worth mentioning that there are only two parameters to be set in this probability machine (a sensitivity factor and a probability threshold). The sensitive factor is a parameter with a clear physical interpretation, which indicates the duration at which the observed probability is 1 predicted probability changes from 0 to 1. This indicates that this probability machine is an easy-to-use method.

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Yan Zhang[a,*], Gunther Notni[a,b]
[a] *Group for Quality Assurance and Industrial Image Processing, Technische Universität Ilmenau, Ilmenau, Germany*
[b] *Fraunhofer Institute for Applied Optics and Precision Engineering IOF Jena, Jena, Germany*

[*] Corresponding author.
E-mail addresses: yan.zhang@tu-ilmenau.de (Y. Zhang), gunther.notni@tu-ilmenau.de (G. Notni).